

Evaluation of a Pharmaceutical Care Program for Hypertensive Patients in Rural Portugal

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Objectives: To evaluate the community pharmacist's capacity to positively influence the results of antihypertensive drug therapy through a pharmaceutical care program and to determine what factors limit the program. **Design:** Randomized, controlled study. **Setting:** Private pharmacy caring for a semiliterate, rural Portuguese population. **Patients:** Random sample of 100 patients with a diagnosis of essential hypertension who had been on drug treatment for less than 6 months. Patients were randomly assigned to an intervention ($n = 50$) or a control ($n = 50$) group. **Intervention:** Individualized health promotion by a research pharmacist involving monthly appointments for 6 months to monitor blood pressure; assess adherence to treatment; prevent, detect, and resolve drug-related problems (DRPs); and encourage nonpharmacologic measures for blood pressure control. Control patients received traditional care. **Main Outcome Measures:** Control of blood pressure; decreases in systolic/diastolic blood pressure; number of detected, resolved, and prevented DRPs. **Results:** From the initial sample of 100 patients, 41 patients in the intervention group and 41 patients in the control group completed the longitudinal study. After 6 months, prevalence of uncontrolled blood pressure decreased by 77.4% in the intervention group ($P < .0001$) and by 10.3% in the control group ($P = .48$). Systolic blood pressure fell from a mean $\pm$ standard deviation of 152 mm Hg $\pm$ 23 mm Hg to 129 $\pm$ 15 mm Hg in intervention patients and 148 $\pm$ 16 mm Hg to 143 $\pm$ 20 mm Hg in control patients ($P < .001$). Twenty-four of 29 (83%) detected actual DRPs were resolved. About 40% of potential DRPs were prevented. **Conclusion:** In this rural community, a pharmaceutical care program was associated with significant improvements in blood pressure control in hypertensive patients.

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Despite an overall improvement in hypertension control and a progressive decrease in cardiovascular mortality in the second half of the 20th century in Australasia, Japan, North America, and Western Europe, the World Health Organization estimates that more than 70% of hypertensive patients in these areas are still grappling with imperfect blood pressure control.^{1,2}

In Portugal, studies have indicated that prevalence of uncontrolled hypertension ranges from 55% to 68% among hypertensive patients.³⁻⁵ These studies involved urban populations, and their results cannot be extrapolated to rural populations, for which there is insufficient information about the prevalence of uncontrolled hypertension. However, von Hafe et al.⁵ noted a trend toward poorer blood pressure control in rural areas, particularly among older women.

Hepler and Strand⁶ defined pharmaceutical care as the respon-

sible provision of drug therapy for the purpose of achieving definite outcomes that improve a patient's quality of life. The outcomes are curing a disease, eliminating or reducing symptoms, arresting or slowing a disease process, and preventing a disease or symptoms. To achieve these ends, pharmacists cooperate with patients and with other health care professionals in designing, implementing, and monitoring a care plan aimed at preventing and resolving drug-related problems (DRPs).⁶

Studies in various countries have demonstrated that the active involvement of a pharmacist on the health care team can improve poor hypertension control and inadequate management of drug therapy.⁷⁻¹² We designed a program that focused on providing pharmaceutical care to hypertensive patients in rural Portugal. Our program planning was guided by the application of pharmaceutical care principles to hypertension control, as defined at the World Health Organization (WHO) Second Meeting on the Role of the Pharmacist.¹³ This strategy entails initiating a health promotion and patient education scheme that specifically targets hypertension and applying a methodology that involves a patient care process oriented toward antihypertensive drug therapy, in order to identify potential and actual DRPs, resolve actual DRPs, and prevent potential DRPs.

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Objectives

The aims of this study were to evaluate the community pharmacist's capacity to positively influence the results of antihypertensive drug therapy through a pharmaceutical care program and to determine what factors limit the program.

The main outcome measure was blood pressure control. Therefore, we tested two null hypotheses:

H₀₁—No final percentage differences between patients who attained blood pressure control in the intervention group (receiving pharmaceutical care) and the control group (receiving standard care).

H₀₂—No final differences in mm Hg between mean variations registered in the intervention group and the control group.

Methods

Study Site

The study was carried out in a private pharmacy that serves a rural area in the western part of Portugal, about 40 km north of Lisbon. The region's 3,000 residents live in 8 small villages. The pharmacy is located in Maxial, which has a population of 700. The public health center, which employs two primary care physicians and one nurse, is located near the pharmacy. The pharmacy employs one pharmacist (the research pharmacist, who is also the pharmacy owner and technical director) and one pharmacy technician. It has an area of 80 m², and pharmacist-patient consultations are held in a private office.

Design and Patient Selection

An unpublished cross-sectional study carried out during February 2000 in a representative sample of 100 patients determined that the prevalence of uncontrolled hypertension in the population the pharmacy serves was 80%.

For the present study, we identified an initial sample of 100 patients from among the pharmacy's patients who were taking prescribed antihypertensive drugs for treatment of essential hypertension, were receiving care from one of the physicians in the public health center, and had their prescriptions filled in the pharmacy. Patients confirmed the diagnosis of primary hypertension. When their diagnosis was uncertain—some patients were using antihypertensive drugs for reasons other than hypertension (e.g., beta-blockers for prevention of migraine)—the research pharmacist contacted the physician. From this initial sample, patients were randomly assigned to either a control group or an intervention group. In this phase we excluded patients who were unable to attend scheduled meetings (bedridden patients), patients who were not interested in participating in the study, and patients who had a spouse enrolled in the study in the other group. To avoid potential bias in the results, we also excluded patients who had been on antihypertensive drug therapy for less than 6 months.

Informed consent was obtained from all patients in the study.

Intervention Group Pharmaceutical Care

Patients in the intervention group were followed for a period of 6 months (from April 2000 through September 2000). Intervention patients had their blood pressure measured during monthly scheduled interviews with the research pharmacist. Each patient's pharmacy record contained sociodemographic data, clinical and therapeutic data, patient behaviors, lifestyle information, and blood pressure records. Where applicable, these data were obtained and updated monthly.

The nonpharmacologic measures were applied to all intervention patients and covered the principal environmental and dietary determinants of blood pressure (i.e., salt and potassium, physical inactivity, obesity, and alcohol). The pharmacist interviewed patients, obtained laboratory and body measurements, and designed an individualized nonpharmacologic care plan on the basis of the results.

Educational leaflets were handed out during the first interview. Oral and written instructions were also given during the first interview and reinforced in the subsequent interviews. The educational materials were of two types: leaflets distributed by the Portuguese National Association of Pharmacies containing general advice to hypertensive patients and specific materials developed by the research pharmacist, for instance, a list of foods with high salt and high cholesterol content and diets for obese hypertensive patients.

The research pharmacist had trained intensively in taking blood pressure readings using an interactive CD-ROM published by the British Hypertension Society,¹⁴ which tests observers in several reading sequences by comparing their answers with correct answers given by experienced observers. A mercury sphygmomanometer with a set of three bladders was used, and measures were taken with the patient seated after 5 minutes of rest, in accordance with the American Heart Association standards and guidelines.¹⁵ The results of two readings were averaged. Baseline and final blood pressure readings were taken at the same time of day at each interview, taking into account circadian blood pressure fluctuations. When patients confirmed they were under unusual stress (nonstudy-related), the blood pressure reading was not recorded.

Blood pressure control was defined as systolic blood pressure (SBP) below 140 mm Hg and diastolic blood pressure (DBP) below 90 mm Hg. These values are considered the lower limits of hypertension according to the *Sixth Report of the Joint National Committee on Prevention, Detection, Evaluation, and Treatment of High Blood Pressure* (JNC VI)¹ and the *WHO Guidelines for the Management of Hypertension*.²

The pharmacist's recommendations for drug regimen changes in light of the DRPs detected were made to physicians mainly by letter but also verbally by telephone. Recommendations to patients focused mainly on nonpharmacologic measures and adherence reinforcement.

For classification of DRPs, we used the *Consensus of Granada*,¹⁶ which divides the three drug-related needs *necessity*, *effectiveness*, and *safety* into two types of DRPs (see Table 1).

Table 1. Classification of Drug-Related Problems According to the Consensus of Granada

Drug-Related Need	Drug-Related Problem	
<i>Necessity</i>	Type 1	Patient does not use the drug he or she needs
	Type 2	Patient uses a drug he or she does not need
<i>Effectiveness</i>	Type 3	Patient does not respond to the drug
	Type 4	Patient uses a dose, interval, or duration inferior to the one needed
<i>Safety</i>	Type 5	Patient uses a dose, interval, or duration superior to the one needed
	Type 6	Patient uses a drug that produces an adverse drug reaction

Source: Reference 16.

Control Group

Control patients received traditional pharmacy services consisting of brief counseling, medication review, and monitoring for adverse drug reactions. Control patients had their blood pressure measured at baseline and after 6 months. The same method for blood pressure measurement was used as for the intervention patients, and these patients contacted the pharmacist only when a medication refill was needed.

Data Analysis

Sample size calculation for the longitudinal study was made using SAMPZISE V2.0.¹⁷

We set α at 0.05 and β at 0.2 (power of 80%) and used bilateral tests to determine that:

Two groups of 20 patients with controlled blood pressure were needed, assuming final differences of 40% between groups in the percentage of patients that could be controlled at the end of the study (main outcome).

Considering the intermediate outcome (blood pressure variations in mm Hg), two groups of 33 patients were needed, assuming final differences of 7 mm Hg DBP between groups.

Statistical analysis was performed using SPSS 10.0 and Epi Info 6 (available at www.cdc.gov/epiinfo/ei6.htm). We analyzed continuous data using Student *t* tests: paired-sample tests for within-group analysis and independent-sample tests for between-group comparisons. Categorical data were analyzed using χ^2 , and Fisher exact test was used when expected cell values were less than 5.

Significance was set at $P = .05$.

Results

From the initial sample of 100 patients, 41 patients in the intervention group and 41 patients in the control group completed the

longitudinal study.

Two patients in each group were excluded at the beginning of the study because their spouses were members of the other group. Two members of the intervention group were excluded because they had used antihypertensive medication for less than 6 months. Five patients in the intervention group were dropped from the study. The reasons for discontinuation were missing more than one scheduled interview ($n = 2$), patient's lack of belief in any health-related added value in the study ($n = 2$), and patient moved out of the area ($n = 1$). Seven patients in the control group were dropped—four for missing the final scheduled interview and three because of repeated difficulties in contacting them. We registered no significant differences between the demographic, clinical, therapeutic, and lifestyle variables of those patients excluded from follow-up and those who completed the study. These variables were comparable for the two groups at baseline (see Table 2).

Blood Pressure Control

The decreases in uncontrolled blood pressure at the end of the study in the intervention group were statistically significant ($\chi^2 = 28.3$; $P < .0001$), whereas decreases in the control group were not ($\chi^2 = 0.50$; $P = .48$).

From 31 intervention patients with uncontrolled blood pressure at baseline, 7 (22.5%) remained uncontrolled at the end, meaning that 24 (77.4%) patients achieved blood pressure control (decrease of 77.4% in uncontrolled blood pressure). In the control group, only 3 of 29 uncontrolled patients at baseline achieved blood pressure control (10.3% decrease) (see Figure 1).

The registered difference of 67.1% between the groups at the end of the study was significant ($\chi^2 = 27.24$; $P < .0001$).

As a result, we rejected the null hypothesis H_{01} .

Blood Pressure Variations

In the intervention group, we noted a significant decrease of 23.15 mm Hg ($P = .0001$) in SBP and 12.34 mm Hg in DBP ($P = .0001$) between the baseline and final means.

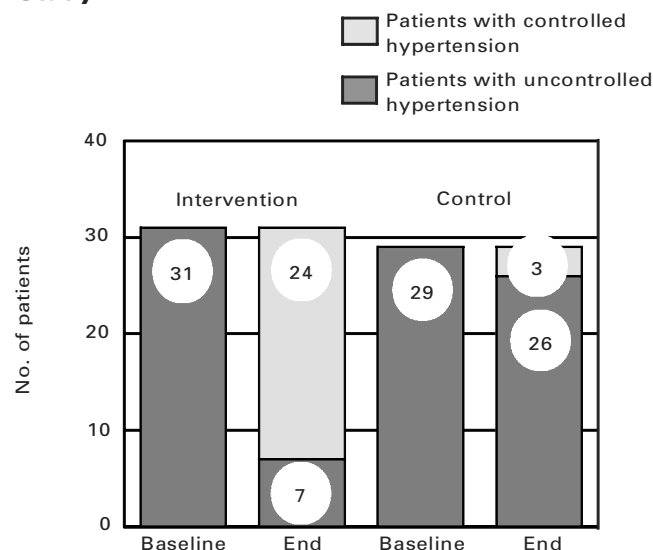
In the control group, we observed a significant decrease of 5.32 mm Hg between baseline and final means for DBP. The decrease in SBP was not statistically significant.

Student *t* tests for independent samples showed a significant difference ($P = .0001$) of 18.36 mm Hg between variations of the means of SBP in the intervention group at the end of the study and a significant difference ($P = .001$) of 7.02 mm Hg in DBP (see Tables 3 and 4).

Therefore, we also rejected the null hypothesis H_{02} .

Drug-Related Problems

We documented a total of 34 DRPs (a prevalence of 59%) in the intervention group, 29 of which were actual DRPs and 5 potential DRPs (see Table 5). From the actual DRPs detected, five corre-

Figure 1. Changes in the Number of Patients With Uncontrolled Hypertension From Baseline to End of Study

sponded to the drug-related need *necessity* and were all type 2. Three of these were therapeutic duplications (two for use of two diuretics with action at the same pharmacologic site of nephron and one for duplication of the same drug). The other two type 2 DRPs had to do with patients with extremely low blood pressure values taking three antihypertensive drugs without a comorbidity that justified each medication.

Sixteen DRPs corresponded to the drug-related need *effectiveness*. Eight of these were type 3 (patient does not respond to the drug). The other eight, all of which resulted from nonadherence, were type 4.

The eight actual DRPs corresponding to the drug-related need *safety* were all type 5. These patients' blood pressure records showed excessive measures, but they were nonetheless prescribed the same dose of the antihypertensive drug at the same frequency in spite of presenting with symptoms of hypotension (e.g., dizziness).

We detected five potential DRPs, all type 6 (also under *safety*). Three were linked to the use of thiazides in high doses (50 mg hydrochlorothiazide) and two were linked to the use of short-acting nifedipine. Potential DRPs were prevented by drug substitu-

Table 2. Homogeneity Between Groups at Baseline

	Intervention Group (n = 41)	Control Group (n = 41)	Statistical Test	P
Demographic variables				
Sex			²	.32
Men, no. (%)	14 (34.1)	9 (22.0)	—	—
Women, no. (%)	27 (65.9)	32 (78.0)	—	—
Age, mean – SD	66.56 – 8.21	63.48 – 12.70	t test	.197
Education level			²	.699
Illiterate, no. (%)	17 (41.4)	14 (34.1)		
Basic schooling, no. (%)	20 (48.8)	21 (51.2)		
Basic schooling +, no. (%)	4 (9.3)	6 (14.7)		
Clinical and therapeutic variables				
SBP, mean – SD (mm Hg)	151.68 – 23.16	147.70 – 15.97	t test	.368
DBP, mean – SD (mm Hg)	85.66 – 13.16	83.90 – 9.18	t test	.486
Patients with uncontrolled hypertension, no. (%)	31 (76.0)	29 (70.73)	²	.622
BMI, mean – SD	27.41 – 3.94	29.85 – 4.51	t test	.13
Waist circumference, mean – SD (cm)	103.95 – 12.34	108.75 – 12.30	t test	.08
No. of antihypertensive agents, mean – SD	1.80 – 0.84	1.61 – 0.13	t test	.28
Angiotensin II receptor blockers, no. (%)	5 (6.9)	7 (9.5)	² or Fisher exact test	> .05
-blockers, no. (%)	5 (6.9)	6 (8.2)	² or Fisher exact test	> .05
Calcium channel blockers, no. (%)	5 (6.9)	12 (16.4)	² or Fisher exact test	> .05
Diuretics, no. (%)	25 (34.7)	23 (31.5)	² or Fisher exact test	> .05
ACEIs, no. (%)	32 (44.4)	25 (34.2)	² or Fisher exact test	> .05
Lifestyle variables				
Diet preoccupations, no. (%)				
None	5 (12.2)	8 (19.5)	² or Fisher exact test	> .05
Yes, mainly for maintaining or losing weight	15 (36.4)	11 (26.8)	² or Fisher exact test	> .05
Yes, mainly for lowering salt intake	21 (51.4)	22 (53.8)	² or Fisher exact test	> .05
Alcohol < 30 mL daily intake (men) / 15 mL (women)	3 (7.3)	4 (9.7)	² or Fisher exact test	> .05
	38 (92.7)	37 (90.3)	² or Fisher exact test	> .05
Physical activity, no. (%)				
None	40 (97.5)	38 (92.6)	² or Fisher exact test	> .05
Occasional	1 (2.5)	1 (2.5)	² or Fisher exact test	> .05
Regular	0	2 (4.9)	² or Fisher exact test	> .05

ACEI = angiotensin-converting enzyme inhibitor; BMI = body mass index; DBP = diastolic blood pressure; SBP = systolic blood pressure; SD = standard deviation.

Table 3. Blood Pressure at Baseline and End of Study

Blood Pressure Readings	Intervention Group Mean – SD	Control Group Mean – SD
Baseline		
DBP, mm Hg	85.66 – 13.16	83.90 – 9.19
SBP, mm Hg	151.68 – 23.16	147.71 – 15.98
Final		
DBP, mm Hg	73.32 – 8.20	78.59 – 8.55
SBP, mm Hg	128.54 – 15.06	142.9 – 20.42

DBP = diastolic blood pressure; SBP = systolic blood pressure; SD = standard deviation.

tion after the research pharmacist made recommendations to the physician.

At the end of the study, 24 of 29 (83%) detected actual DRPs were resolved and 2 of 5 (40%) potential DRPs were prevented.

Some actual and potential DRPs remained unresolved or were not prevented. The resolution and prevention of these DRPs required modifications in drug regimens that depended exclusively on a medical decision. While the pharmacist made recommendations in these cases, in the 6-month time frame of the study, physicians did not agree to implement any change to the antihypertensive drug therapy regimen.

Discussion

In our opinion, blood pressure control characterizes the results of pharmaceutical care on hypertensive patients better than blood

pressure decreases since, theoretically, overzealousness of the researcher may exaggerate falls in blood pressure values in intervention patients, actually leading to a worse quality of life in this study group. Nevertheless, there is a considerable amount of literature that deals more emphatically with quantitative outcomes (i.e., mm Hg decreases as a result of pharmaceutical care practice).^{6–9,11} One reason for this may be that sample size, when calculating differences in proportions, would necessarily be larger than when continuous mean differences are involved. In the present study (compared with other studies), detecting, resolving, and preventing DRPs was of central importance since, in pharmaceutical care practice, they affect morbidity and mortality.¹⁸ Also, we tried to apply appropriate *t* tests to compare poststudy measures between the intervention and control groups, rather than, as other studies have done, using simple calculations of mean differences from baseline, consequently hindering the interpretation of the findings.

We observed significant poststudy differences between groups, both in blood pressure control (67.1%) and mm Hg decreases (18.36 SBP and 7.02 DBP). The improvements in the intervention group, compared with control group, do not appear to be due to regression to the mean. The extent of the decreases, both in mm Hg and in uncontrolled blood pressure, which were also observed in the control group, may have been inflated because of seasonal variations in blood pressure. Such seasonal variations have been reported in studies involving a large number of patients (e.g., the WHO MONICA Project¹⁹). Also, for ethical reasons, the research pharmacist resolved some DRPs in the control group, an action that may have contributed to this decrease in blood pressure. This intervention contaminated the control group, introducing some bias.

An initial high prevalence of uncontrolled blood pressure (80%) in the study population could be predictive of large final differ-

Table 4. Results of Independent *t* Tests

	Mean	SE Mean	Lower	Upper	<i>t</i>	<i>df</i>	<i>P</i> (Two-Tailed Test)
SBP differences (intervention and control)	18.36	3.24	9.86	26.8	4.3	80	.0001
DBP differences	7.02	1.65	2.89	11.15	3.3	80	.001

DBP = diastolic blood pressure; *df* = degrees of freedom; SBP = systolic blood pressure; SE = standard error.

Table 5. Distribution of Actual and Potential Drug-Related Problems

Drug-Related Need	DRP Type	Actual DRPs		Potential DRPs		Total No. (%)
		Resolved	Unresolved	Prevented	Not Prevented	
<i>Necessity</i>	1					
	2	4	1			5 (14.7)
<i>Effectiveness</i>	3	6	2			8 (23.5)
	4	6	2			8 (23.5)
<i>Safety</i>	5	8				8 (23.5)
	6			2	3	5 (14.8)
Total no. (%)	—	24 (82.7)	5 (17.3)	2 (40.0)	3 (60.0)	34 (100)

DRP = drug-related problem.

ences between groups regarding blood pressure control (40%) and mm Hg (an effect size of 0.7). The final number of 82 patients (two groups of 41) who completed the study comfortably exceeded the sample size of 66 patients (two groups of 33) needed to have a power of 80%.

Since pharmaceutical care is a new concept in Portugal and its practice is just starting to interest some professionals, as far as we are aware at this time (December 2000), this is the first randomized, controlled study done in Portugal that used a methodology centered on identifying, resolving, or preventing DRPs. Even in the excellent studies carried out abroad (mainly in the United States), it is not often, regarding both the hypertension issue and a pharmacist's involvement in team care, that researchers emphasize what many consider the core of this new professional practice—identifying, resolving, or preventing DRPs.

A close interprofessional relationship between health care providers, which is fundamental to pharmaceutical care practice and is the privilege of those working in a small rural setting, facilitated communication in this project.

Again, the aim of this study was to evaluate the community pharmacist's capacity to positively influence the results of antihypertensive drug therapy through a pharmaceutical care program. Our results—better blood pressure control and significant decreases in SBP and DBP in the intervention group—show that this was achieved. A major factor in achieving this was probably the resolution of 83% of all DRPs detected. Determining specifically which pharmacist-implemented approach (nonpharmacologic measures versus the DRPs approach) led to this outcome was not within the scope of this study, but it is certainly an interesting topic for future research.

Limitations

Because the research pharmacist assigned patients to the two groups, the study was only single-blinded. This could have introduced bias. In addition, the research pharmacist took all blood pressure readings; measurements made by patients, physicians, or nurses were not part of the study. We believed that measurements by other providers would have introduced further bias because of the lack of a standardized, reliable blood pressure measurement technique and the use of different devices.

Because of the potential seasonal variations in blood pressure, the study should have lasted for a full year, but for practical reasons, that length of time was not possible.

The relatively small number of patients in the study sample (82 patients) allowed us to detect significant final differences in blood pressure values and uncontrolled hypertension between groups. The high prevalence of uncontrolled blood pressure detected in the sample used in the previous cross-sectional study indicated the strong possibility of detectable final differences, which was valuable information for sample size calculation. If researchers plan to detect smaller differences than the ones detected in this study, in the same

rural setting or in an urban setting in populations with better blood pressure control, larger size samples would naturally be needed.

We explored only clinical outcomes in this study. We did not follow up to investigate humanistic outcomes because of a lack of extensive significant results from other studies also focusing on pharmaceutical care for patients with hypertension and using samples of similar size.^{8,9,12}

The researchers' lack of experience in pharmaceutical care may have prevented the detection of more DRPs in the intervention group. These undetected DRPs may have caused worse clinical results. Assuming that this new practice had to be learned from scratch, as it represented a significant break with traditional practice, a longer training period for the researchers might have produced better results.

Because this study was carried out in a small rural setting with patients with poor blood pressure control, our results may not be generalizable to urban populations in Portugal or elsewhere.

Cutoff points other than 140/90 mm Hg for blood pressure control were not considered for statistical calculation (i.e., blood pressure goal of < 130/85 mm Hg for higher-risk hypertensives with either diabetes mellitus or established renal impairment).^{1,2} However, these cutoff points were taken into account during the follow-up. In this manner, we managed to bring blood pressure down to the desired level, according to the criteria stated above, in five of the seven patients with diabetes whose blood pressure was uncontrolled at baseline.

Conclusion

In this randomized, controlled trial in a rural community pharmacy, pharmaceutical care practice was centered on detecting, resolving, and preventing DRPs and reinforcing nonpharmacologic measures. Our results show that applying the pharmaceutical care philosophy to hypertensive patients in a rural setting can help in the control of these patients' blood pressure, and consequently lower the risk that hypertension represents for cardiovascular disease.

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— A P h A S E S Q U I C E N T E N N I A L : V O I C E S F R O M P A S T I S S U E S O F J A P h A —

Attitudes Toward Treating Hypertension Change

Careful use of drug and nondrug therapies can reduce blood pressure in most elderly hypertensive patients without causing significant side effects, according to a report sponsored by the American Heart Association (*JAMA*. 1990;264:1015-8).

The authors wrote that until recently, hypertension was considered a normal result of aging. A change in attitude about treating elderly hypertensives has been prompted by controlled studies showing that therapy reduces cardiovascular morbidity and mortality in elderly persons.

Elderly patients should undergo a thorough evaluation before therapy begins, the authors state. This should include taking blood pressure readings with the patient lying down and after he or she has been standing for one to two minutes.

Recommended nondrug treatments are the same as for younger patients: losing weight if obese, lowering sodium intake, reducing alcohol consumption, and getting regular aerobic exercise.

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